

Adjoint Matrix

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Learning Outcomes

At the end of this lecture, you should be able to:

- Know what is an adjoint matrix and its properties.
- Find the inverse of the matrix by its adjoint.
- Use the Cramer's rule to solve the system of linear equations.

Adjoint matrix

Let $A = [a_{ij}] \in M_n(\mathbb{R})$. The **adjoint** of A , denoted by **adj** A , is the transpose matrix of cofactors of the element of a_{ij} .

$\text{adj}A = [A_{ij}]^T$ where A_{ij} is the cofactor of a_{ij} in $\det A$,

$$\text{adj}A = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}.$$

Adjoint matrix

Let A be an $n \times n$ matrix.

- 1 $\text{adj}A$ must exist.
- 2 $\text{adj}A$ is an $n \times n$ matrix.
- 3 $\text{adj}A$ is unique.
- 4 $\text{adj}A$ is the transpose of the matrix of cofactors of a_{ij} .

Adjoint matrix

Example:

1. Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Find $\text{adj}A$.

Properties of the adjoint

Let A be an $n \times n$ matrix. Then

$$(1) A(\operatorname{adj} A) = |A|I_n = (\operatorname{adj} A)A$$

$$(2) |\operatorname{adj} A| = |A|^{n-1} \text{ if } |A| \neq 0$$

Properties of the adjoint

Let A and B be an $n \times n$ matrices. Then

$$(1) \operatorname{adj}(A + B) \neq \operatorname{adj}A + \operatorname{adj}B$$

$$(2) \operatorname{adj}(AB) = (\operatorname{adj}B)(\operatorname{adj}A)$$

$$(3) \operatorname{adj}(A^{-1}) = (\operatorname{adj}A)^{-1} \text{ if } A \text{ is invertible}$$

$$(4) \operatorname{adj}(A^T) = (\operatorname{adj}A)^T$$

$$(5) \operatorname{adj}(\lambda A) = \lambda^{n-1}(\operatorname{adj}A) \text{ for non-zero } \lambda \in \mathbb{F}$$

Properties of the adjoint

Let A be an $n \times n$ matrix. Then A is invertible if and only if $\text{adj}A$ is invertible.

Furthermore, if A is invertible, then $A^{-1} = \frac{\text{adj}A}{|A|}$.

Example:

Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Is A invertible? If so, find A^{-1} by using the adjoint of A .

Cramer's rule

Cramer's Rule:

Let A be an $n \times n$ invertible matrix. Then the system of linear equations $AX = B$ where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}, \text{ has a}$$

unique solution and the unique solution of the system is given by $x_i = \frac{\det A_i}{\det A}$, where, for each $1 \leq i \leq n$, let A_i be the matrix obtained from A by replacing the i^{th} column of A by the column matrix B , i.e.,

$$\det A_i = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1,i-1} & b_1 & a_{1,i+1} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2,i-1} & b_2 & a_{2,i+1} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{n,i-1} & b_n & a_{n,i+1} & \cdots & a_{nn} \end{vmatrix}$$

Cramer's rule

Example:

Solve the following system of linear equations using Cramer's Rule:

$$x + y + z = 6,$$

$$x + 2y + 3z = 14,$$

$$x + 4y + 9z = 36.$$

Cramer's rule